VARUN MANDEPUDI

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EDUCATION

Master of Science in Data Science Sep 2025 - May 2027
Northeastern University, Khoury College of Computer Sciences, Boston, MA *(GPA: 3.50/4.00)*
Relevant Courses: Machine Learning, Database Management Systems, Essentials of Data Science, Algorithms

B. Tech in Computer Science with Specialization in Big Data Analytics Sep 2021 – May 2025
SRM University, Chennai, India *(GPA: 3.73/4.00)*
Relevant Courses: Data Science, Business Intelligence and Analytics, Machine Learning, Probability and Queuing Theory

TECHNICAL SKILLS

Languages & Tools: Python (PyTorch, Scikit-Learn, Pandas, NumPy), R, SQL, Bash, Git/GitHub, Jupyter Notebooks
AI / Machine Learning: Deep Learning, Neural Networks (CNNs, LSTMs, Transformers), NLP, RAG Pipelines, Model Evaluation, Data Cleaning, Cross-Attention
Fusion Analytics: Statistical Analysis, A/B Testing, Regression, EDA, Forecasting, Time-Series Analysis, Feature Engineering
Cloud & Data: Google Cloud (Vertex AI), Azure Databricks, PySpark, PostgreSQL, MySQL, FAISS, Flask APIs
Visualization: Power BI, Tableau, Excel, ggplot2, Matplotlib, Seaborn

WORK EXPERIENCE

Data Analytics Intern – Saint-Gobain India, Chennai, India Aug 2026 - Sep 2026
- Built a physics-based Python framework for screening low-emissivity PVD glass coatings, combining transfer-matrix optics, Fuchs–Sondheimer transport, and multi-objective desirability scoring to Pareto-rank coating stacks
- Executed MACE-MP-0 and CHGNet interatomic potential surrogates on Google Cloud Vertex AI, identifying a composition-substitution route to ~82% silver mass reduction under ECBC 2017 constraints
- Replicated an ICLR 2025 inverse molecular design model on Vertex AI, adding 30+ tests and a manifest verifier that surfaced 18 data discrepancies; presented findings in an hour-long technical review

Data Scientist Intern – Fragiliti Narratives, Bangalore, India Jun 2026 - Jul 2026
- Built business-facing data flows and data stories on the company's analytics platform, translating raw operational data into narratives for non-technical decision-makers
- Verified Monte Carlo simulation and forecasting algorithms by reproducing platform outputs in Python, generating 1,500+ rows of synthetic fixtures via Gaussian copula sampling tuned within 0.004 of target correlations
- Designed a 17-case edge-condition suite (zero-variance, boundary, invalid inputs) that surfaced a defective precondition allowing a positive semi-definite check to pass unconditionally, and traced a NaN propagation path to its source

Undergraduate Researcher – SRM University, Chennai, India Oct 2024 - Apr 2025
- Waste Classification Using ML Models: A Comparative Study — Benchmarked CNN, ResNet, and classical ML architectures for automated image-based waste classification; accepted and presented at ICRDICCT'25 (Apr 2025 DOI: 10.5220/0013927500004919
- Transformative Potential of AI and ML in the Energy Sector: Smart Grids — Proposed ensemble deep learning approaches for smart grid optimization and energy demand forecasting; published at Com-IT-Con 2024 (Oct 2024) DOI: 10.1201/9781003650010-144

Data Analyst Intern – Thaai Casting Limited Sep 2024 - Nov 2024
- Built Python data pipelines to clean and process 50,000+ production and sales records, reducing data inconsistencies by ~30% for downstream ML analysis
- Applied clustering (K-Means) and classification models on regional performance data across 10+ regions, contributing to an 8% increase in regional sales
- Developed 5+ interactive Power BI dashboards tracking 15+ manufacturing KPIs, enabling data-driven decision-making for 3 stakeholder teams

PROJECTS

RAG Chatbot for Research Papers | Python, PostgreSQL, FAISS, Flask, Mistral/Ollama
- Engineered a semantic search pipeline over 547,000+ arXiv papers using sentence-transformers and FAISS, achieving sub-second retrieval in a 768-dimensional embedding space
- Built a PostgreSQL-backed vector store with optimized FAISS indexing, cutting query latency ~40% versus brute-force search.
- Integrated Mistral LLM in a retrieval-augmented generation framework returning cited source papers

Smart Lecture Classifier| Python, PyTorch, ViT-B/16, BERT, Cross-Attention, Hugging Face
- Developed a multimodal classifier for 4-class pedagogical categorization of lecture segments using ViT-B/16 (vision) and BERT (text) with cross-attention fusion
- Aligned visual and textual lecture data with stratified splits and macro-F1 evaluation; ablation studies showed ~12–15% accuracy gain from multimodal fusion over unimodal baselines